ANDREW DE LA VARA

Grand Junction, CO | (720) 610-4770 | andrewdelavara571@gmail.com | linkedin.com/in/andrewdelavara | ande3836.github.io

TECHNICAL SKILLS

CAD & Analysis: SolidWorks (modeling, assemblies, technical drawings, GD&T, sheet metal), MATLAB, Navisworks
Construction & Project Tools: Bluebeam, Procore, Viewpoint, Egnyte, SharePoint, Microsoft Project
Office & Data: Microsoft Excel, Word, PowerPoint, OneDrive, Teams
Fabrication & Engineering: Extensive 3D printing, prototype iteration, quantity takeoffs, RFIs, submittals, budget estimates, scoping
Certifications & Languages: CDOT Flagger (2026–2028), BLS Provider (2025–2027); English (native), Spanish (full professional)

EDUCATION

University of Colorado Boulder **Boulder, CO**
Bachelor of Science in Mechanical Engineering; GPA: 3.2 Expected May 2027
- Completed coursework through Heat Transfer and Thermodynamics 2; transferred from Colorado Mesa University in Fall 2025
- **Honors:** National Hispanic Scholar (2024, 2026); 3rd Place, Academic Olympiad, SHPE National Convention (2026)

EXPERIENCE

MTech Mechanical, Mountain Team **Eagle, CO**
Project Engineer September 2026 – Present
- Perform **Bluebeam** quantity takeoffs of domestic water, waste/vent, hydronic piping, and ductwork to estimate labor hours
- Developing an **automation tool** that imports Bluebeam **Polylength** pipe lengths into a labor spreadsheet to predict install hours

Project Engineering Intern June 2026 – August 2026
- Contributed to about one-fourth of the **150 RFIs** on the **$7 million** Frisco Transit Center Headquarters and created submittals
- Aggregated **50+ vendor documents** for BABAA and FTA compliance; managed files in Procore, Egnyte, SharePoint, Viewpoint
- Scheduled **final inspections** for State Boiler, mechanical, and plumbing systems during closeout and supported procurement

Ridgeview Data Collection **Grand Junction, CO**
Field Technician May 2025 – Present
- Collected **traffic speed and volume data** for **CDOT, Colorado Mesa University, Colorado Parks and Wildlife**, and others
- Set up and take down field equipment alone or in a team; receive, upload, and format data in OneDrive and Teams

Nellas Frozen Yogurts **Littleton, CO**
Shift Manager May 2022 – June 2024
- Led a team of **4–7 employees** each day and resolved customer complaints and requests 4–5 times a week

PROJECTS

Western Filament Senior Design **Boulder, CO**
Project Team Lead Fall 2026 – Present
- Lead a team designing a vertically integrated line that combines coating and multiwinding to raise efficiency by **25%**
- Created the **project scope and budget estimates** and presented the **preliminary design review** to our sponsor for approval
- Iterated a proof-of-concept mini winder, selecting a **self-reversing traverse** for far fewer components and lower cost

Frictionless Motion Air Track, Colorado Mesa University Physics Department **Grand Junction, CO**
Project Team Lead January – May 2025
- Led the design of a **20-part SolidWorks assembly** and worked with **Recla Metals** to bend the track into shape
- Designed and **3D printed** the air distribution and track end cap system, resulting in a **fully functioning model**

Parkinson's Walking Assist Belt
- Conducted **extensive market research** to identify a gap for an **affordable walking assist device** for Parkinson's patients
- Developed a prototype belt that uses **visual cueing** to help Parkinson's patients walk more fluidly

LEADERSHIP & ACTIVITIES

Society of Hispanic Professional Engineers (SHPE), Colorado Mesa University Chapter **Grand Junction, CO**
Fundraising Officer
- Plan fundraisers, including a Pie-a-Professor event with engineering faculty, to fund member convention travel

Colorado Mesa University Varsity Football, Student Athlete (3 seasons) **2023 – 2025**